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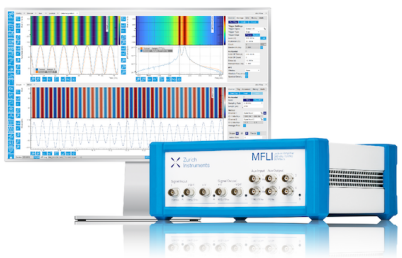
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Beyond Born–Oppenheimer spectroscopic study for the *C* state of LiH

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A theoretical spectroscopic study is performed for the *C* state of LiH using eight diabatic or adiabatic states of symmetry $^1\Sigma$ beyond the Born–Oppenheimer approximation. For the position of the vibrational levels, an excellent agreement is found with the ten recently observed ones giving confidence to the present study. The vibrational levels present rather strong vibronic shifts (rising to about 10 cm^{-1} for $v=37$) particularly for the highest ones. Two approximate methods are tested for their estimation and a new method is presented for the evaluation of the vibronic part of the adiabatic correction. The new method involves the diabatic picture and the discrete variable representation. It leads to nonlocal corrections but avoids the calculation of radial couplings. Nonradiative lifetimes are accurately determined. They range over six orders of magnitude, from nanoseconds to milliseconds. Similarities of the LiH molecule in the *C* state and in higher excited states at large internuclear distances with highly excited Rydberg atoms are emphasized. © *1999 American Institute of Physics.* [S0021-9606(99)01423-3]

I. INTRODUCTION

The lithium hydride molecule has attracted a large interest in recent years. It is the simplest heteroatomic molecule and presents various spectroscopic interests. An avoided crossing between an ionic (Li^+H^-) and a neutral [$Li(2s)H(1s)$] electronic configuration imprints the ground (X) and the first excited (A) states of symmetry $^1\Sigma$. The resulting A state is flat bottomed and highly anharmonic. These two states have been intensively studied and their spectroscopy is rather well characterized (for a review see Ref. 1). Because of the small reduced mass, non-Born–Oppenheimer (BO) effects plays here a noticeable role despite the largely avoided crossing. Nonradiative lifetimes and vibronic shifts have been computed recently for all the vibrational levels of the A state.[2,3]

The second excited state of $^1\Sigma$ symmetry, labeled C, presents even more unusual characteristics with an interesting double well shape.[4] The main well is for rather large internuclear distances (equilibrium distance about 10.2 a.u.) while the other one is for higher energies and much shorter distances (about 3.8 a.u.). Very recently the C state has been observed[5] for the first time using two laser techniques. Ten vibrational levels were observed[5] and tentatively assigned with the help of previous theoretical predictions.[4]

In addition to these intriguing spectroscopic characteristics, the C state as well as the other highly excited states of LiH present various interesting particularities. They all present a main well which extends over a very wide range of internuclear distances and an outer ionic branch associated with huge permanent dipoles.[6] For the C state, the permanent dipole rises up to about 18 a.u., while values as large as 31, 34, and 92 a.u. are attained for the D, E, and F states, respectively. These maxima being reached for similar values of the internuclear distance (i.e., about 20.0, 32.5, 35.5, and 92.4 a.u. for the C, D, E, and F states, respectively). Extremely small Franck–Condon factors are expected because of such large distances which makes the direct excitations from the ground state very difficult. However we suggest that these highly excited states may present very interesting properties as molecular analogues of Rydberg atoms. In collisions with atomic or molecular species the cross-section should be large because of the large internuclear distances involved and also because of the large dipoles. In addition, the density of rovibrational states is high. Thus, these excited states may be used as tunable precursors in collision-induced resonant energy transfers allowing for some new features in experimental and theoretical spectroscopy. Highly excited Rydberg atomic states are actually used in this way and tuned with electric fields.[7,8] We have here both electronic and vibrational degrees of freedom which could bring interesting new possibilities. The hydrogen anion (H^-) behaves, for the molecule in the ionic configuration, as the Rydberg electron in atoms. This analogy may be further exploited. The detailed study performed here for the C state could be useful for that purpose. However, there is an obvious but important difference between the atomic and the molecular situation: a Rydberg electron has a single internal state while H^- may loose an electron (which jumps on the Li^+) and the excited molecule dissociates. Indeed, nonadiabatic effects should be expected to play an important role in such a perspective.

The large distance neutral-ionic crossing which imprints

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the bottom of the main well is less avoided than the one occurring for the *A* state. Enhanced non-Born–Oppenheimer effects are thus expected. An objective of the present contribution is to estimate them accurately. Following our previous study for the *A* state, we will concentrate on the vibronic shifts and the nonradiative lifetimes. For the evaluation of the vibronic shift, a test is performed for two approximate methods: the recent improved van Vleck's correction and the usual adiabatic correction for which a new approach using a discrete variable representation (DVR) is presented. Taking benefit of the diabatic representation determined in a previous work, we improve the ionic curve and compare the results with the recently observed ones.

The presentation is organized as follows. In Sec. II the theory used for the evaluation of the various nonadiabatic effects (adiabatic correction, full vibronic shift, improved van Vleck's estimation, and nonradiative lifetime) is presented. The results are reported and discussed in Sec. III. Concluding remarks are given in Sec. IV.

II. THEORY

The present study involves a set of eight electronic states in either the diabatic or the adiabatic representations and a DVR grid[9–12] for the vibrational basis set. In the diabatic representation (ϕ_j) which was determined previously by an *ab initio* calculation[4,6] we have only potential couplings. However these couplings may be rather large, in particular for the short interatomic distances. These couplings are better taken into account in the adiabatic representation in which the electronic states (ψ_i) are coupled by differential operators. The two electronic representations are connected by an *R*-dependent unitary transformation

$$|\psi_i\rangle = \sum_j U_{ij}(R)\,|\phi_j\rangle. \tag{1}$$

The matrix $\mathbf{U}(R)$ diagonalizes the diabatic electronic Hamiltonian matrix $\mathbf{V}(R)$ (i.e., the potential matrix also called effective Hamiltonian matrix) which depends parametrically on the internuclear distance *R*:

$$\mathbf{U}^{\dagger}(R)\mathbf{V}(R)\mathbf{U}(R) = \mathbf{e}(R), \tag{2}$$

where $\mathbf{e}(R)$ is the diagonal matrix corresponding to the usual adiabatic *ab initio* energies.

Since the matrix $\mathbf{U}(R)$ obviously depends on *R*, the adiabatic states are coupled by the vibrational kinetic energy operator *T* which in the diatomic nonrotating case reduces to

$$T = -\frac{1}{2\mu}\frac{\partial^2}{\partial R^2}, \tag{3}$$

where μ is the usual reduced mass of the diatomic molecule.

The eigenvalues and eigenvectors of the full Hamiltonian can be obtained from the diagonalization of the full vibronic matrix (operator $T+V$) in either the diabatic or the adiabatic representation.

A. Diabatic representation

Using the diabatic representation, the full Hamiltonian matrix in the DVR is readily written in the form:

$$H_{ikjl} = T_{kl}\delta_{ij} + V_{ij}(R_k)\,\delta_{kl}. \tag{4}$$

In the entire paper we adopt the convention that *i* and *j* are electronic state indices while *k* and *l* are vibrational basis set ones.

Interestingly, in the DVR the potential energies (V_{ii}) and potential couplings (V_{ij}) are diagonal. Moreover, assuming strictly diabatic states, the kinetic matrix (T_{kl}) does not depend on the electronic states and vanishes by construction for different diabatic states i,j ($i \neq j$). Therefore, the full matrix is rather sparse. We used here an equidistantly spaced DVR which involves a Fourier transformation for the evaluation of the kinetic matrix according to the so-called fourrier grid Hamiltonian (FGH) method.[13] The *C* state presents an unusual double-well shape and potential-optimized DVR[14] are not very useful.

We are interested here in vibronic states which are above the *X* and *A* asymptotes, hence, they are embedded in the two continua. The diagonalization provides simultaneously the vibronic resonances we look for, and also continuum states discretized by the finite vibrational basis set. A stabilization procedure is necessary to identify the resonances. They have large bound amplitudes and therefore are not very sensitive to variations of the DVR parameters. For example by varying the parameter $R_{\max}$, the resonances energies are easily identified as the stable eigenvalues. The stabilization also allows for an estimation of the accuracy of the calculation since various densities of DVR points are experimented. Under the conditions specified in the next sections, the resonance energies are accurate to at least 10^{-3} cm^{-1}.

From the stabilization diagram, the nonradiative lifetimes could be in principle derived according to the method proposed by Hazi and Taylor.[15] However, important numerical difficulties arise. As for the *A* state, the nonradiative lifetimes are of the order of the nanosecond which corresponds to widths of the order of 10^{-6} cm^{-1}, a value three orders-of-magnitude smaller than the typical accuracy of the DVR. The coupling between the bound states and the continua is too small to be derived from the stabilization diagram. An alternative could be the accurate calculation of continuum functions as in the close coupling approach. We choose here to keep the DVR but to revert to an approach involving the complex plane and Green function techniques with the help of an imaginary optical potential.

B. Adiabatic representation

The full Hamiltonian matrix($\mathbf{H^a}$) presents now various terms,

$$H^a_{ikjl} = T_{kl}\delta_{ij} + e_i(R_k)\;\delta_{ij}\delta_{kl} + W_{ikjl}. \tag{5}$$

The last term matrix ($\mathbf{W}$) is neglected in the Born–Oppenheimer approximation.[16] Vibrational levels can then be determined for each adiabatic potential independently of the others. We will refer to these energies as the adiabatic ones that will be noted E^a. Beyond the Born–Oppenheimer approximation, this term is responsible for two important effects: the vibrational levels present (i) a finite nonradiative

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lifetime due to nonadiabatic transitions to states in the continua, and (ii) a shift in energy which can be rather large as will be seen later.

1. Usual approach for the adiabatic correction

Usually W is expressed as a local R-dependent operator which couples the adiabatic states:

$$W_{ij}(R) = -\frac{1}{2\mu}\left[h_{ij}(R) + 2f_{ij}(R)\frac{\partial}{\partial R}\right], \tag{6}$$

where

$$f_{ij} = \langle \psi_i | \frac{\partial}{\partial R} \psi_j \rangle, \tag{7}$$

and

$$h_{ij} = \langle \psi_i | \frac{\partial^2}{\partial R^2} \psi_j \rangle. \tag{8}$$

f_{ij} can be easily computed from the diabatic representation using a Hellman–Feymann expression

$$f_{ij}(R) = \frac{1}{e_j - e_i}\left[\mathbf{U}^{\dagger}(R)\frac{\partial \mathbf{V}}{\partial R}\mathbf{U}(R)\right]_{ij}. \tag{9}$$

Then h is simply evaluated using the closure approximation of the electronic representation

$$h_{ij}(R) = [f^2(R)]_{ij} + \frac{\partial f_{ij}(R)}{\partial R}. \tag{10}$$

Only h presents nonvanishing diagonal elements which lead to a repulsive contribution $-1/2\mu\, h_{ii}(R)$ when they are added to the adiabatic energies. This is the so-called adiabatic correction. From these corrected adiabatic curves, neglecting all the other nonadiabatic terms, new vibrational levels can be determined which will be noted E^{ac}. As in the preceding publication devoted to the X and A states of LiH,[2] in the present work we only evaluate that part of the adiabatic correction which is directly related to the diabatic–adiabatic transformation. For a complete evaluation, others' geometry dependent contributions[17] due to nuclear translation and rotation should be calculated using *ab initio* techniques.[18–20] For the ground state of LiH (X), experimentally inferred adiabatic corrections could be derived from detailed spectroscopic data.[21,22] However these studies rest upon a single-state approach. We are interested here in a comparison between the full vibronic shift and its single-state approximation which reduces to the preceding contribution.

The preceding approach only requires the knowledge of $V_{ij}(R)$ from which all couplings (f and h) can be determined. However care has to be taken with respect to the consistency of the phase factors in the matrices **U** for the various distances.

2. New approach for the adiabatic correction

Taking better advantage of the DVR another approach can be developed which completely avoids the evaluation of the couplings f and h. Since the unitary transformation matrix **U** is required in the DVR only for a set of discrete values of R (the grid points R_k), let us define a global unitary transformation as the algebraic sum of these local rotation matrices,

$$U_{ikjl} = U_{ij}(R_k)\,\delta_{kl}. \tag{11}$$

This global rotation transforms the full Hamiltonian matrix **H** into a matrix $\tilde{\mathbf{H}}^{\mathbf{a}}$ similar to the $\mathbf{H}^{\mathbf{a}}$ one. It is interesting now to compare these two matrices.

$$\tilde{\mathbf{H}}^{\mathbf{a}} = \mathbf{U}^{\dagger}\mathbf{H}\mathbf{U} = \tilde{\mathbf{T}}^{\mathbf{a}} + \tilde{\mathbf{V}}^{\mathbf{a}}. \tag{12}$$

Obviously, this transformation fully diagonalizes the potential matrix

$$\tilde{V}^{a}_{ikjl} = e_i(R_k)\,\delta_{ij}\,\delta_{kl}. \tag{13}$$

Similarly, one could expect that the matrices $\mathbf{T}^{\mathbf{a}} + \mathbf{W}^{\mathbf{a}}$ and $\tilde{\mathbf{T}}^{\mathbf{a}}$ are identical. This is not the case. Because of the particular structure of the matrices **U** and **T**, the matrix $\tilde{\mathbf{T}}^{\mathbf{a}}$ presents a rather simple expression

$$\tilde{T}^{a}_{ikjl} = T_{kl}\sum_m U^{\dagger}_{im}(R_k)U_{mj}(R_l). \tag{14}$$

We thus arrive at a surprising result: $\tilde{\mathbf{T}}^{\mathbf{a}}$ and $\mathbf{T}^{\mathbf{a}}$ have identical diagonal matrix elements (since for the diagonal $R_k = R_l$ and the matrices $\mathbf{U}(R)$ are unitary). Apparently there is now no adiabatic correction. In fact the adiabatic correction now has a nonlocal character. It is spread out from the diagonal to the off-diagonal intrablock matrix elements (a block corresponds to a given electronic state). Interestingly, the diagonalization of the submatrices $\tilde{\mathbf{H}}^{\mathbf{a}}_{\mathbf{ii}}$, which corresponds to the standard neglect of the interstate coupling $\tilde{\mathbf{T}}^{\mathbf{a}}_{\mathbf{ij}}(i \neq j)$ as in the usual adiabatic correction approach, leads to the usual E^{ac} energies [where $-1/2\mu\, h_{ii}(R)$ was added to the potential and all the $\mathbf{W}_{\mathbf{ij}}(i \neq j)$ terms were neglected]. This approach presents the extremely convenient aspect that no phase problem arises and no operator matrix elements have to be evaluated. This is the interesting advantage of directly using the representation of the kinetic operator within the DVR instead of its analytic expression. There is now no numerical derivative to perform but products of small rotation matrices; the job is done intrinsically by Eq. (14). This approach is interesting because in many cases the adiabatic correction gives a correct estimation of the full vibronic shift. Only the reduced matrices corresponding to the given electronic states (a given block of the full vibronic matrix) needs here to be diagonalized which can be useful when several coordinates are involved.

The new method profits from performing calculations in the diabatic basis. Interestingly, in the *ab initio* determination of the diabatic basis and potentials, the calculation of the radial couplings was also avoided here[6,4,23] as well as in various diabatization methods.[24] The exciting point now is that, while completely avoiding the calculation of radial couplings from the *ab initio* level to the dynamics, important nonadiabatic effects such as the adiabatic correction can be evaluated by combining appropriate diabatization method with DVR approach.

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3. Improved van Vleck's correction

Recently an improved van Vleck's correction formula was proposed[25] for the evaluation of nonadiabatic energy shifts. This proposal goes beyond the adiabatic correction, the additional effect of the interstate nonadiabatic coupling being estimated in a simple way. The formula involves the radial coupling matrix elements $f_{ij}(R)$ which indeed needs to be determined previously. Here also, we used a Hellman–Feynmann approach together with a phase-fixer algorithm.[2] For a vibrational state v (function $|\chi_{iv}\rangle$) of energy E_{iv} corresponding to the ith electronic state, the contribution of the jth electronic state to the additional shift is estimated as[25]

$$\Delta E_{iv}^{(j)} = \frac{1}{\mu^2}\langle\chi_{iv}| \frac{2\mu(E_{iv}-e_j(R))f_{ij}^2(R)+\left(\partial f_{ij}(R)/\partial R\right)^2}{e_i(R)-e_j(R)}|\chi_{iv}\rangle. \tag{15}$$

The total additional shift is the sum of the various contributions

$$\Delta E_{iv}=\sum_j \Delta E_{iv}^{(j)}. \tag{16}$$

In our application, the eight lowest adiabatic states were involved. This additional correction takes into account only the interstate nonadiabatic couplings. In order to get a correction similar to the full vibronic shift, the usual adiabatic corrections (intrastate) were added to ΔE_{iv}.

C. Nonradiative lifetime

As for the A state, the vibrational levels of the C state have rather large nonradiative lifetimes which makes difficult their accurate computation. We used here an approach which was shown to be reliable for the A state.[3,26] It combines a Feshbach partition technique for the Green operator, the use of an optical potential that performs the required analytical continuation of the functions into the complex plane, and a perturbative expansion. The higher order terms were calculated in order to stabilize the results with respect to the optical potential. Remarkably stable results were found for the A state,[3] all in an excellent agreement with the ones of the close coupling,[2] and even when badly chosen optical potentials were tried.[3] We briefly give below an outline of the method.[3]

According to the Feshbach formalism, two orthogonal projectors are defined. The projector P corresponds to an approximated description of the resonant state while the projector Q is associated with the orthogonal complement in the Hilbert space:

$$P=|\varphi\rangle\langle\varphi|, \quad Q=1-P. \tag{17}$$

Then we have to determine the pole of the total resolvent $G(z)=(z-H)^{-1}$ that corresponds to the P-space state. It can be shown (see for example Ref. 27) that its position is obtained by solving the transcendental equation

$$z=\langle\varphi|H+H\frac{Q}{z-H}H|\varphi\rangle=\langle\varphi|H+HG_Q(z)H|\varphi\rangle. \tag{18}$$

However, resonances are associated with z values in the lower complex halfplane in which the partial resolvent $G_Q(z)$ must be analytically continued. This is done by adding an optical potential V^{opt} to the Hamiltonian[28–30]

$$H^{\text{opt}}=H-i\lambda V^{\text{opt}}. \tag{19}$$

Since the resonance energies must not depend on the optical potential, the influence of V^{opt} is progressively reduced by means of perturbation theory. This leads to the final equation for the resonance energy z_R (the lifetime is half the inverse of the imaginary part) at the nth order of perturbation

$$z_R^{(n)}=\langle\varphi|H|\varphi\rangle + \sum_{k=0}^{n}\langle\varphi|H[G_Q^{\text{opt}}(z_R^{(n)})i\lambda V^{\text{opt}}]^k G_Q^{\text{opt}}(z_R^{(n)})H|\varphi\rangle, \tag{20}$$

where $G_Q^{\text{opt}}=Q/(z-H^{\text{opt}})$. The equation may be iterated from an initial guess, e.g., $z_R^{(0)}=\langle\varphi|H|\varphi\rangle$. Correct results are produced when the expansion stabilizes in a few orders of perturbation, typically about $n=5$.

III. RESULTS AND DISCUSSION

We used the diabatic *ab initio* results $[V_{ij}(R)]$ calculated over a wide range of internuclear distances from $R=1.8$ a.u. to $R=243$. a.u. As reported in previous studies,[4,6,31] this *ab initio* data leads to an excellent agreement with the experimental results for the spectroscopy of the X and the A states as well as for the neutralization cross-section at low energies.[32] The present calculation involves the seven lowest-lying $^1\Sigma$ neutral channels $[\text{Li}(2s,2p,3s,3p,3d,4s,4p)\text{H}(1s)]$ and the ionic one (Li^+H^-). The matrix elements are interpolated on the DVR grid using cubic splines. Several calculations have been performed using various DVR grids: (i) usual grids with 200 to 250 points and varying $R_{\max}$ from 25 to 45 a.u. for the nonradiative lifetimes, (ii) medium grids with 500 points and $R_{\max}$ varying between 50 and 80 a.u. for the stabilization approach to identify the resonances and compute the full vibronic shift, and (iii) fine grids with 2000 points for the adiabatic (single state) Born–Oppenheimer calculations and also for the improved van Vleck's and the adiabatic correction evaluation of the vibronic shift. In all cases, in order to insure consistency between the adiabatic and the diabatic schemes, a particular extrapolation procedure to very short distances is applied, as described in Ref. 2.

A. Diabatic and adiabatic *ab initio* potentials

Before going further with the vibronic results it is worthwhile to examine the nature of the electronic states involved in the C state according to the diabatic and the adiabatic representations. The potentials are displayed in Figs. 1 and 2 for the adiabatic $[e_i(R)]$ and the diabatic $[V_{ii}(R)]$ schemes, respectively.

In order to improve the *ab initio* results, a correction is brought specifically to the ionic curve taking benefit of the diabatic representation. This correction accounts for basis set limitation in the *ab initio* full CI calculation of the hydrogen

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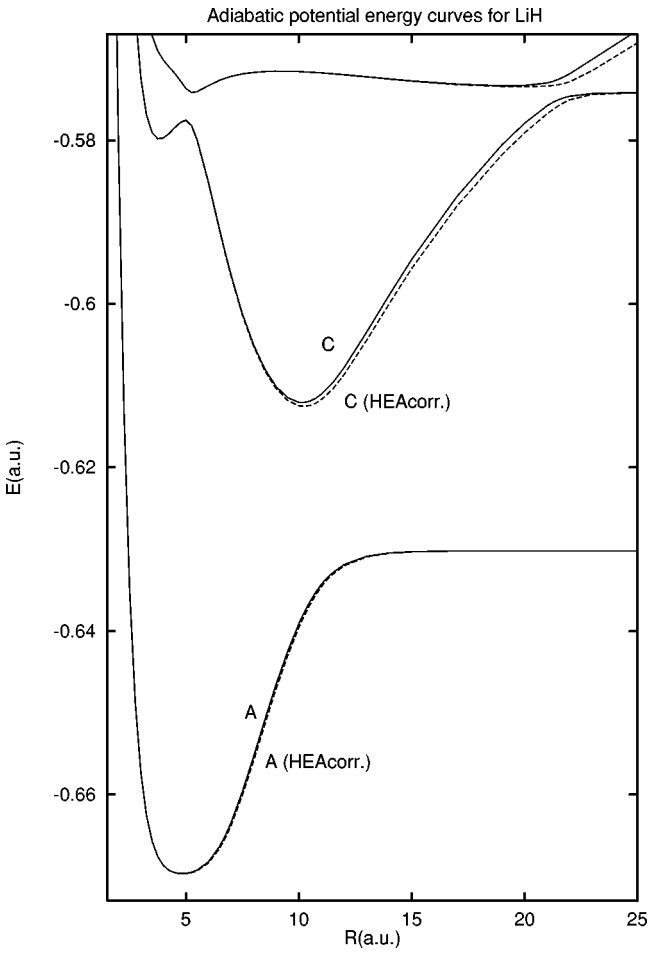


FIG. 1. *Ab initio* adiabatic potential energy curves for the $C^1\Sigma$ state and its neighbors. Full lines: direct *ab initio* results; dotted lines: improved results including a correction of the hydrogen electron affinity (HEAcorr).

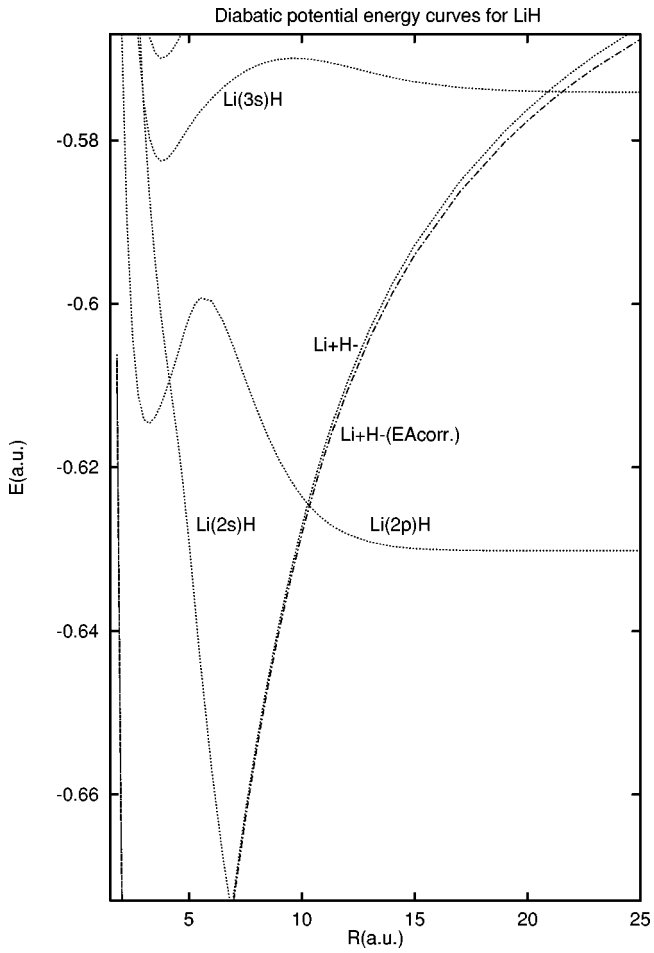


FIG. 2. As Fig. 1 but for the diabatic curves most involved in the *C* state. Only the diabatic ionic curve is modified by the HEA correction.

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electron affinity. The main problem arises from the calculation of the H^- anion which requires huge basis sets with high angular momentum functions in order to take into account accurately the electronic correlation. The hydrogen electron affinity (HEA) is accurately known both from theoretical and experimental determinations, HEA=0.754 209 eV,[33,34] while we have at infinite separation HEA=0.703 883 eV. However, as the lithium atom approaches, its basis set improves the local completeness around the hydrogen atom and the error decreases. An *R*-dependent correction has thus been determined[4] for the ionic diabatic potential specifically. As can be seen in Fig. 2, the corrected ionic potential (Li^+H^- HEAcorr) is slightly shifted to lower energies with respect to the direct *ab initio* result (Li^+H^-). Since the correction is performed only for the ionic diabatic potential, changes in the adiabatic potentials reveal ionic components of the adiabatic wave functions. In Fig. 1 we see that only the large distance branch of the main well is affected and therefore is clearly ionic. The correction induces a global increase of the width of the main well. As for the *A* state, a general decrease of the vibrational level spacing is then expected.

The diabatic picture (Fig. 2) reveals an intricate electronic structure for the *C* state which results from a series of crossings. From large distance to short ones, it is imprinted successively by a $Li(3s)H(1s)$ asymptote, an ionic branch, a repulsive neutral $Li(2p)H(1s)$ branch, a very repulsive $Li(2s)H(1s)$ branch, and finally we have again the $Li(3s)H(1s)$ branch. The surprising shape of the diabatic states has been related[4] to intrinsic characteristics of the corresponding Rydberg orbitals. The triplet diabatic states present very similar shapes[4] excluding therefore an alternative interpretation involving potential mixing with other ionic states.

Although for large distances (typically *R* larger than 7 a.u.) the adiabatic state is dominated by a single diabatic state (or locally, around an avoided crossing, a couple of them), for short distances electronic couplings are large and the imprint of the diabatic states is less evident. However the various crossings contribute to peaks in the radial coupling,[35] the neutral–ionic ones at *R* about 22 a.u. and *R* about 10 a.u., and the neutral–neutral ones at shorter distances. The latter peaks have been confirmed by direct *ab initio* calculations of the radial coupling.[36] They could be related to the orbital promotions predicted by Barat and Lichten.[37] The physical insight of these short distance crossings can be related to the spatial characteristics of the orbitals involved. The valence Li $(2s)$ orbital is much less diffuse than the Rydberg ones and therefore more repulsive effects are expected when it is

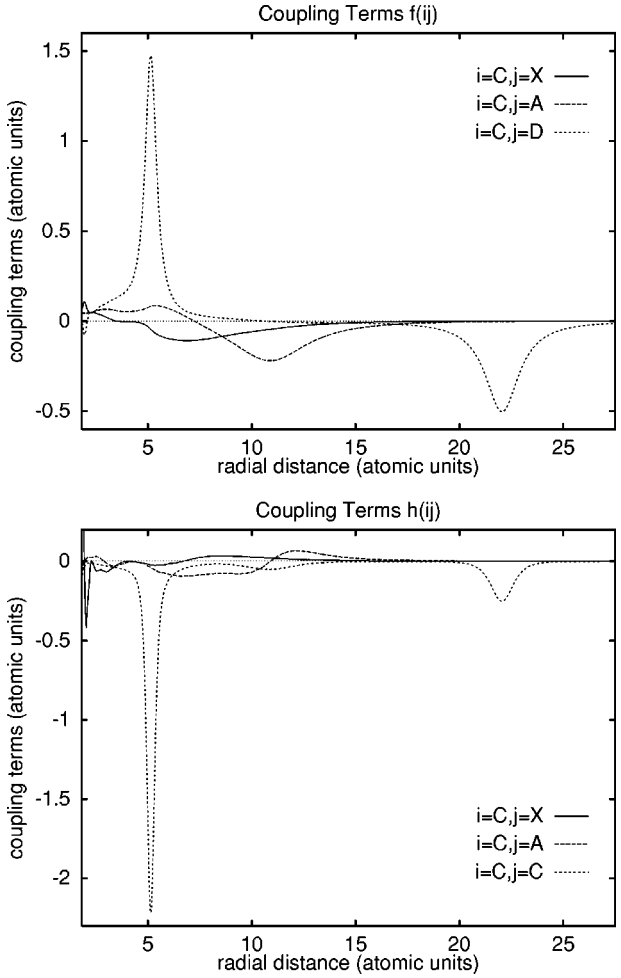


FIG. 3. Nonadiabatic coupling terms $f_{ij}=\langle\psi_i|\partial/\partial R|\psi_j\rangle$ (a) and $h_{ij}=\langle\psi_i|\partial^2/\partial R^2|\psi_j\rangle$ (b) involved in the golden rule estimation of the nonradiative lifetime (f_{CX}, f_{CA}, $h_{CX}$and h_{CA}) and in the adiabatic correction (h_{CC}). f_{CD} is also depicted to show the origin of the peak in h_{CC}.

interpenetrated by the localized H(1s) orbital. The overrepulsiveness of the Li$(2s)$H$(1s)$with respect to the other neutral configurations leads to a series of crossings in the diabatic picture. These crossings become largely avoided and completely hidden in the adiabatic approach, however they contribute effectively to the nonadiabatic effects. Such crossings should also play an important role for inelastic process in atomic collisions, especially at high energy collisions. The matrix elements f_{CX}, f_{CA}, and f_{CD} calculated using Eq. (9) are displayed in Fig. 3(a) and h_{CX}, h_{CA}, and h_{CC} [Eq. (10)] in Fig. 3(b). For the C state, h_{CC} contributes to the adiabatic correction while the interstate couplings f and h contribute to the nonradiative decay and to the vibronic shift of the vibrational levels.

As emphasized in Ref. 6 the ionic branch of the adiabatic wells is associated with a large permanent dipole moment which behaves as $c_{ion}R$ (c_{ion} being the ionic component of the adiabatic wave function). For the C state, c_{ion} is about 1 in the window from $R=10$ to $R=20$ a.u. where the dipole rises to about 20 a.u. Very intense infrared spectra are thus expected, in particular for the higher vibrational levels.

B. Stabilization of the resonances

To identify the resonances of the C state we varied the extension of the DVR grid (R_{max}) while keeping fixed its number of points to 500. The resonances have a large bound amplitude and therefore are not very sensitive to variations of the DVR parameters while the continuum states do. The full vibronic matrix of dimension 4000 (8 electronic states $\times$ 500 DVR points) is constructed using the diabatic representation and diagonalized for four values of the parameter R_{max} (50, 60, 70, and 80 a.u.). The resonances are then easily identified as the stable eigenvalues in between the bottom of the well and the Li$(3s)$H$(1s)$ asymptote. The stabilization also allows for an estimation of the accuracy in the calculation since the density of points vary by a factor of nearly 2. The resonance energies are accurate to better than 10^{-3} cm^{-1}.

The position of the resonances corresponding to the vibrational levels of the C state and also their spacing are reported in Table I for the *ab initio* potentials and for the improved ones (HEA corrected) together with the recent experimental results. The position of the resonances were computed with respect to the X asymptote. For this comparison, we add the experimental dissociation energy (D_e) of the ground state ($D_e=20\,288$ cm^{-1}) (Ref. 1) instead of the theoretical one which has already a small error ($D_e=19\,994$ cm^{-1} for the *ab initio* data and $D_e=20\,149$ cm^{-1} for the improved *ab initio* data).

The vibrational level spacing for $v=12$ to 20 is in excellent agreement with the experimental observations and the first observed level can be safely assigned to $v=12$. As anticipated, with the HEA correction we get a general decrease of all spacings: $\Delta E_v^u>\Delta E_v$ for all v (ΔE_v^u corresponds to the unmodified *ab initio* data and ΔE_v to the improved one). As for the A state, it is remarkable that all the experimental spacing are bracketed by the theoretical results with and without the correction.

$$\Delta E^u>\Delta E^{exp}>\Delta E, \tag{21}$$

an additional indication that the ionic branches of the wells correspond to the main difficulty in the *ab initio* calculation.

We notice that the experimental spacings are somewhat closer to the uncorrected theoretical ones than to the HEA corrected ones, suggesting that, in this energy range, the shape of the well is better described by the direct *ab initio* approach or alternatively that the R-dependence of the HEA correction could be improved.

On the other hand, this correction is of paramount importance for the energy position of the levels. As can be seen in Table I, the energy terms are in excellent agreement with the observations when the HEA correction is taken into account. The error in the theoretical terms ranges from an overestimation by 6 cm^{-1} for $v=12$ to an underestimation by 15 cm^{-1} for $v=21$, a remarkable accuracy for transitions larger than 42 000 cm^{-1} which strongly confirms the assignation of the lowest observed level to $v=12$. The agreement is somewhat less good for the direct *ab initio* results where we have a uniform overestimation of the terms by more than 150 cm^{-1} which clearly shows the importance of the HEA correction.

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TABLE I. Theoretical results in comparison with recent experimental data.[a] E are the exact theoretical energies, E^u the uncorrected ones (i.e., without the correction for the electroaffinity of hydrogen atom), and $E^{\exp}$ the experimental values. Energies are given in cm^-1 with respect to the minimum of the X state potential. For this comparison we used the experimental value of the X state potential depth $D_e = 20\,287.7$. The Δ values represent the spacing between two neighbor levels v and $v+1$.

v	E^u	E	$E^{\exp}$	ΔE^u	ΔE	$\Delta E^{\exp}$
0	39 316.14	39 211.71		292.25	288.07	
1	39 608.39	39 499.78		287.64	283.60	
2	39 896.02	39 783.38		282.99	279.06	
3	40 179.01	40 062.44		278.17	274.45	
4	40 457.18	40 336.88		273.35	269.76	
5	40 730.54	40 606.64		268.52	265.09	
6	40 999.06	40 871.73		263.73	260.40	
7	41 262.79	41 132.13		258.95	255.78	
8	41 521.74	41 387.91		254.21	251.14	
9	41 775.95	41 639.05		249.50	246.56	
10	42 025.45	41 885.61		244.83	241.98	
11	42 270.28	42 127.59		240.23	237.48	
12	42 510.51	42 365.07	42 358.90	235.68	233.01	234.65
13	42 746.19	42 598.08	42 593.55	231.20	228.61	230.93
14	42 977.39	42 826.70	42 824.48	226.77	224.27	226.09
15	43 204.16	43 050.97	43 050.57	222.41	219.98	221.94
16	43 426.56	43 270.95	43 272.51	218.10	215.76	217.37
17	43 644.67	43 486.71	43 489.88	213.86	211.59	213.26
18	43 858.53	43 698.29	43 703.14	209.67	207.47	208.91
19	44 068.20	43 905.77	43 912.05	205.54	203.41	204.86
20	44 273.74	44 109.18	44 116.91	201.46	199.41	200.58
21	44 475.20	44 308.59	44 317.59	197.44	195.46	
22	44 672.64	44 504.05		193.47	191.57	
23	44 866.11	44 695.63		189.55	187.74	
24	45 055.66	44 883.36		185.67	183.94	
25	45 241.33	45 067.30		181.81	180.19	
26	45 423.13	45 247.49		177.97	176.46	
27	45 601.10	45 423.95		174.11	172.75	
28	45 775.21	45 596.70		170.27	169.06	
29	45 945.48	45 765.76		166.31	165.35	
30	46 111.79	45 931.11		162.19	161.58	
31	46 273.98	46 092.69		157.39	157.68	
32	46 431.36	46 250.37		84.39	153.31	
33	46 515.75	46 403.69		70.67	106.89	
34	46 586.42	46 510.58		145.75	44.80	
35	46 732.17	46 555.38		133.34	142 .19	
36	46 865.51	46 697.57		98.08	133.56	
37	46 963.59	46 831.13		91.40	106.45	
38	47 054.98	46 937.58		117.95	84.04	
39	47 172.93	47 021.62		117.08	110.72	
40	47 290.01	47 132.35		103.48	116.71	
41	47 393.49	47 249.06		78.68	107.66	
42	47 472.17	47 356.72			89.60	
43		47 446.32			51.03	
44		47 497.35				

[a]Reference 5.

In Fig. 1 it is readily seen that the main well of the C state is made deeper and wider by the HEA correction. Its main effect is then a general decrease in the position of the vibrational levels and also in their spacing.

In summary the comparison with the experimental results shows that the HEA correction is satisfactory for the term levels where it leads to an accuracy better than 1 for 1000, while the improvement is less satisfactory for the level spacing. We are probably reaching here the limit of the accuracy of the operatorial core–valence correlation estimation which is more important for the ionic state where the two electrons are localized at the hydrogen atom than for the neutral states for which more diffuse orbitals are involved. Therefore the C state of LiH offers a challenging test for the calculation of long-range core–valence correlation.

C. Vibronic shift and nonradiative lifetime

We will now consider for almost all the vibrational levels of the C state two important non Born–Oppenheimer effects, the vibronic shift and the nonradiative lifetime. For this study we used mainly the improved *ab initio* data (with the HEA correction) since it is the most accurate one.

1. Vibronic shift

In addition to the computation of the full vibronic shift determined by a comparison between the full vibronic (exact) energies E_v and the adiabatic BO ones E_v^a, we tested the quality of two approximations: the usual adiabatic correction (AC) approach and the improved van Vleck's (IvV) one.[25] Both avoid the diagonalization of the full vibronic matrix and allow for independent treatment of the adiabatic electronic states, neglecting their nonadiabatic couplings in AC and estimating their effect through a second-order perturbative expansion in the IvV case. The results for the vibronic shifts are reported in Table II. We observe two behaviors.

From $v=0$ to $v=31$ the vibronic shift is rather uniform and slowly increases from 3 to 4.2 cm^{-1}. The AC approximation systematically underestimates the shift although giving its correct order of magnitude while the IvV formula leads to an accurate estimation within a few percent. Here, as for the A state,[31] the following inequality holds

$$E^a < E^{ac} < E. \tag{22}$$

Beyond $v=32$ the vibronic shift presents a rather erratic behavior and the quality of the approximate estimations deteriorates considerably. Changes of sign in the shift are observed for the higher levels, which are poorly reproduced by the approximate approaches. In addition, the magnitude of the shift increases, rising to 9.7 cm^{-1} for $v=37$(about 10% of the spacing) or to -5.5 cm^{-1} for $v=43$. We have here large vibronic effects which result from a delicate balance of repulsive (adiabatic correction, interaction with states lower in energy) and attractive (interaction with states higher in energy) nonadiabatic interactions, and their estimation is difficult. This is the case for $v=38$, for example, where the shift is underestimated by more than 26 cm^{-1} with the IvV approach and overestimated by more than 9 cm^{-1} with the AC one.

It is clear that vibronic couplings cannot be ignored in the latter case ($v>31$) where the shifts are large and change sign from one vibrational state to another. Comparison with experimental results could not rely on the usual adiabatic BO approach. In particular, the validity of the usual RKR derivation of an ''experimental potential'' is questionable since a basic assumption is to neglect the nonadiabatic couplings between the underlying electronic adiabatic states.

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TABLE II. Various approximations of the full vibronic shift for the LiH $^1\Sigma$ C state and nonradiative lifetimes. E is the exact (full vibronic) energy and E^a the adiabatic (Born–Oppenheimer) energy. Δ stays for the full vibronic shift, Δ^{ac} means the adiabatic correction shift and Δ^{vV} is the sum of Δ^{ac} and the improved van Vleck's correction. τ_R is the accurate nonradiative resonance lifetime and τ_F its Fermi golden rule estimation. The units are cm^{-1} for energy and μs for time.

v	E	E^a	Δ	Δ^{vV}	Δ^{ac}	τ_R	τ_F
0	18 924.009	18 921.005	3.004	3.022	2.859	6.8045	7.1926
1	19 212.077	19 209.044	3.034	3.052	2.631	3.7757	3.7266
2	19 495.679	19 492.636	3.044	3.056	2.412	1.2459	1.2057
3	19 774.735	19 771.687	3.048	3.055	2.220	0.8546	0.7993
4	20 049.183	20 046.132	3.051	3.053	2.054	0.7180	0.6551
5	20 318.938	20 315.885	3.053	3.052	1.913	0.7166	0.6298
6	20 584.029	20 580.972	3.056	3.053	1.794	0.7699	0.6532
7	20 844.432	20 841.370	3.061	3.056	1.695	1.0229	0.8239
8	21 100.213	21 097.147	3.067	3.059	1.611	1.6290	1.2423
9	21 351.350	21 348.279	3.071	3.063	1.541	4.7689	3.2034
10	21 597.907	21 594.831	3.076	3.066	1.483	698.4047	753.3131
11	21 839.886	21 836.806	3.080	3.068	1.434	2.0920	2.1965
12	22 077.366	22 074.283	3.083	3.071	1.394	0.4191	0.3878
13	22 310.381	22 307.295	3.086	3.072	1.362	0.1521	0.1349
14	22 538.995	22 535.907	3.089	3.075	1.337	0.0728	0.0632
15	22 763.266	22 760.178	3.088	3.074	1.317	0.0413	0.0352
16	22 983.248	22 980.159	3.089	3.075	1.304	0.0267	0.0223
17	23 199.007	23 195.918	3.089	3.074	1.295	0.0192	0.0160
18	23 410.593	23 407.504	3.090	3.076	1.293	0.0151	0.0125
19	23 618.067	23 614.978	3.089	3.076	1.295	0.0129	0.0107
20	23 821.480	23 818.391	3.088	3.080	1.305	0.0123	0.0101
21	24 020.889	24 017.802	3.088	3.084	1.320	0.0128	0.0105
22	24 216.351	24 213.263	3.089	3.094	1.343	0.0148	0.0125
23	24 407.926	24 404.832	3.094	3.111	1.376	0.0223	0.0180
24	24 595.661	24 592.560	3.100	3.136	1.421	0.0454	0.0361
25	24 779.602	24 776.489	3.113	3.177	1.485	0.2257	0.1699
26	24 959.788	24 956.654	3.134	3.234	1.573	1.0226	1.0287
27	25 136.249	25 133.078	3.171	3.311	1.697	0.0501	0.0430
28	25 309.000	25 305.775	3.226	3.408	1.873	0.0152	0.0127
29	25 478.058	25 474.740	3.318	3.511	2.128	0.0073	0.0061
30	25 643.407	25 639.939	3.468	3.594	2.497	0.0045	0.0037
31	25 804.992	25 801.273	3.719	3.604	3.045	0.0033	0.0027
32	25 962.671	25 958.504	4.167	3.452	3.884	0.0028	0.0023
33	26 115.986	26 110.802	5.183	3.087	5.353	0.0029	0.0025
34	26 222.880	26 215.908	6.971	4.715	8.567	0.9059	2.6002
35	26 267.682	26 264.153	3.528	0.733	5.257	0.0035	0.0031
36	26 409.873	26 403.288	6.585	-1.538	8.989	0.0066	0.0074
37	26 543.433	26 533.845	9.589	-7.666	14.101	0.0955	29.8091
38	26 649.883	26 643.047	6.836	-19.669	17.794	0.0072	0.0039
39	26 733.924	26 738.782	-4.858	-10.310	8.040	0.0045	0.0031
40	26 844.648	26 848.414	-3.767	-7.850	6.623	0.0124	0.0386
41	26 961.355	26 962.951	-1.596	-16.102	10.246	0.0213	0.0134
42	27 069.018	27 071.262	-2.243	-24.180	12.913	0.0048	0.0029
43	27 158.620	27 164.137	-5.517	-21.114	11.174	—	—
44	27 209.652	27 211.362	-1.709	-0.474	0.594	—	—

2. Nonradiative lifetime

The nonradiative lifetime is estimated accurately using a single optical potential of the form R^8 combined with high-order perturbative expansion. From $v=0$ to $v=42$ we get stable results with less than ten orders of perturbation. To check the reliability of the calculation we varied the DVR characteristics and performed systematically various calculations with 200, 225, and 250 grid points, and $R_{\max}$ ranging from 25 to 45 a.u. The results reported in Table II are stable to better than 1% (exept for $v=10$). We could not get reliable results for the highest levels $v=43$ and $v=44$. In addition we performed a usual golden rule estimation. Although this is an approximate method, since the resonances are here well isolated, it is expected to give rather good results. The comparison of the two results allows for confidence in the calculation.

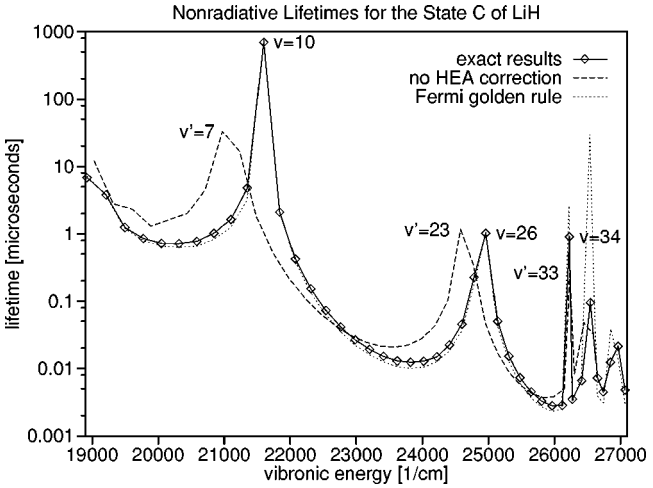


FIG. 4. Nonradiative lifetime of the vibrational levels of the C state as a function of the energy level. Full line: exact calculation using the improved *ab initio* data (HEA correction included); dotted line: golden rule estimation using the same improved *ab initio* data; dashed line: exact calculation using the unmodified *ab initio* data.

As is readily seen in Table II the nonradiative lifetimes vary by six orders-of-magnitude, from a maximum about 700 μs for $v=10$ to a minimum about 3 ns for various levels near the asymptote. The largest nonradiative lifetimes appear as singularities around $v=10$, 26, 34 and to a lesser extent around $v=41$, overimposed on a general decrease as v increases. This behavior is visualized in Fig. 4. It is not an unusual behavior since using semiclassical approaches the peaks could be related to interference effects.[38] As for the vibronic shift, up to $v=33$, we observe a reasonable agreement between the exact and the approximate methods, then the agreement deteriorates due to intricate vibronic effects.

In order to estimate the sensitivity of these results with respect to changes in the potentials, two additional calculations have been performed: one using the golden rule and the potentials including the adiabatic correction (changes of the order of 10^0-10^1 cm^{-1}), the other is an exact calculation but using the *ab initio* potentials without the HEA correction (changes of the order of 10^2 cm^{-1}). In the former case we found insignificant differences with the results of Table II, in the second case quite large differences were obtained. However, as illustrated in Fig. 4, the general behavior is quite similar. The spikes remain but concern neighbor levels. The general structure is thus a robust one but the assignation of the spikes to particular levels is quite sensitive to the potentials and appeals for experimental developments.

The level $v=34$ corresponds to a wave function mainly located in the small upper well around $R=3.8$ a.u. and behaves from all aspects (vibronic shift, level spacing, and nonradiative lifetime) differently than the other levels which correspond essentially to the main well. Of course, for the levels higher than $v=34$, strong interactions between the two wells

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take place, and delocalization also. The levels $v=34$, $v=38$, and some others higher in energy present an interesting particularity, their Franck–Condon factor with the ground-state is strongly enhanced. They are therefore good candidates for direct excitation.

Partial contributions of A and X states to the total nonradiative lifetime can be easily evaluated from the golden rule approach. As could have been anticipated from the smaller energy gap, the contribution of the A state strongly dominates, the one of the X state being rather marginal.

The radiative lifetime is expected to be rather uniform and of the order of some tenth of ns similarly as for the A state.[1] In contrast with the A state where the nonradiative lifetime was nearly negligible compared to the radiative one, it seems to be here of comparable magnitude and even probably smaller than the radiative lifetime for most of the vibrational levels of the C state. The nonradiative lifetime of most of the vibrational levels of the C state could thus be measured rather easily.

IV. CONCLUSION

We have performed a theoretical spectroscopic study of the C state of the LiH molecule and considered with attention various vibronic effects beyond the Born–Oppenheimer approach. The excellent agreement between our theoretical results and the recently observed ones for the term position of the vibrational levels and for their spacing gives a strong support to the present study.

The particular shape of the C state and the large nonadiabatic couplings with others electronic states lead to important non-Born–Oppenheimer effects which call for careful use of the usual inversion methods, specially for the highest vibrational states ($v>31$). Most nonradiative lifetimes are comparable to the usual radiative lifetimes and therefore could be measured experimentally. Despite their large variations, from a fraction of ms to a few ns, the nonradiative lifetimes have been determined accurately using Green function techniques combined with high-order perturbative expansions which remove the influence of the specific optical potential employed.

These unusual spectroscopic properties in which vibronic coupling plays an important role, although unexpected, combined with others unusual characteristics such as large interatomic distances (large radius of the molecule) and huge permanent dipole moments which should lead to intense infrared spectra and large collision cross sections, make this system particularly attractive for new developments both in experiment and theory. The important imprint of the ionic species Li^+H^- and its similarity with a Rydberg atomic system $Atom^+e^-$ could be also further exploited.

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